# SVM - 1

Support Vector Machines ✓

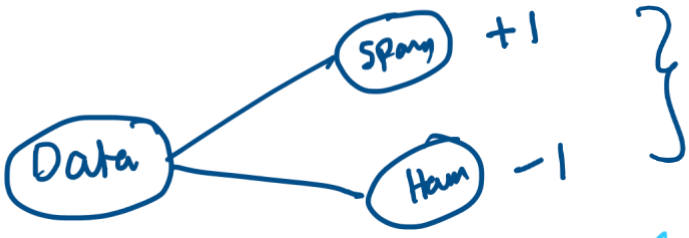
"1990" ✓ "2000" ✓

2008 – 2012 ✓

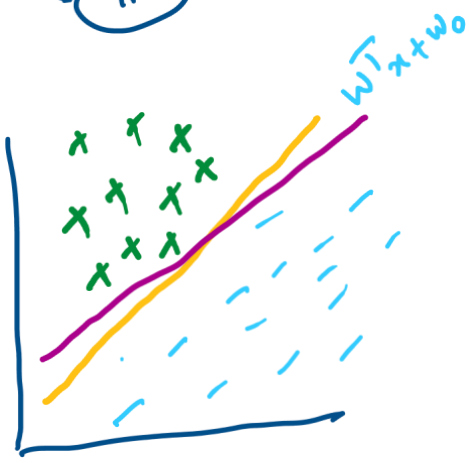
2010 –

"Spam Vs Ham"

Google - Gmail Spam filter.




Inspiration (SVM)

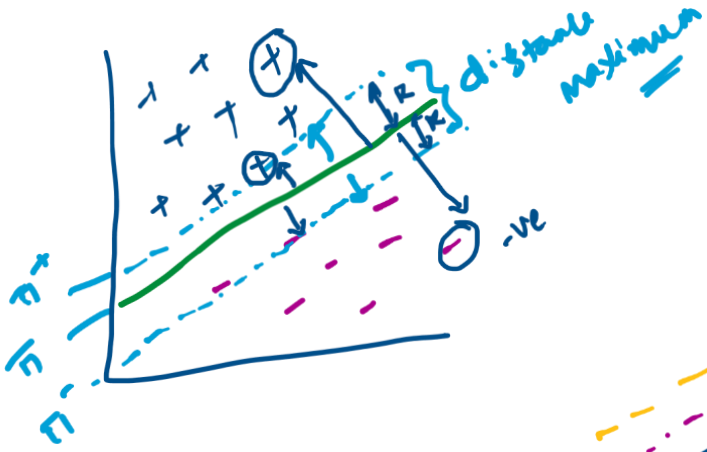



$w^T . x + b$

Maximum Margin

Flexible boundary

Maximum Margin Classifier



$\rightarrow \pi \rightarrow \quad w^T \cdot x + b = 0$

-ve boundary $\pi^- \rightarrow \quad w^T \cdot x + (b + K) = 0$ — ①

+ve boundary $\pi^+ \rightarrow \quad w^T \cdot x + (b - K) = 0$ — ②

$$\frac{b_1 - b_2}{\|w\|} = \frac{(b+K) - (b-K)}{\|w\|} = \frac{2K}{\|w\|}$$

$$\text{Margin}(\pi^-, \pi^+) \rightarrow d(\pi^-, \pi^+) \sim \frac{2K}{\|w\|}$$

**Case 1**

$k = 1$

$$d = \frac{2 \times 1}{\|w\|} \; ?$$

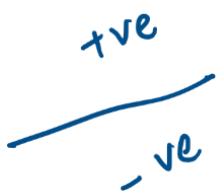
**Case 2**

$k = 10$

$$d = \frac{2 \times 10}{\|w\|} \rightarrow \; w$$

for mathematical Simplicity $k = 1$ ✓

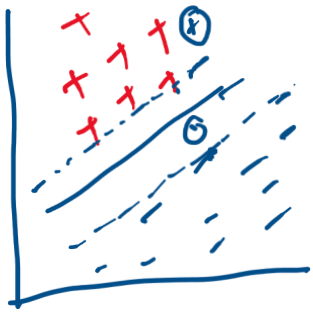
$$\left.\begin{array}{ll} \pi^+ & w^T.x + b - 1 = 0 \\ \pi^- & w^T.x + b + 1 = 0 \end{array}\right\} \; d(\pi^+, \pi^-) = \frac{2}{\|w\|}$$




$$w^T.x + b = 1$$

$$w^T.x + b = -1$$

ML

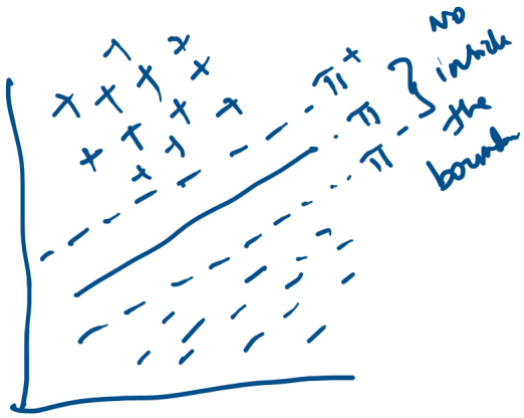


**Generalization**

perform well unseen data

$$\frac{c_1 - c_2}{d\|w\|} = \frac{(\cancel{6} + 50) - (\cancel{6} - 40)}{\|w\|} = \frac{90}{\|w\|} \checkmark$$

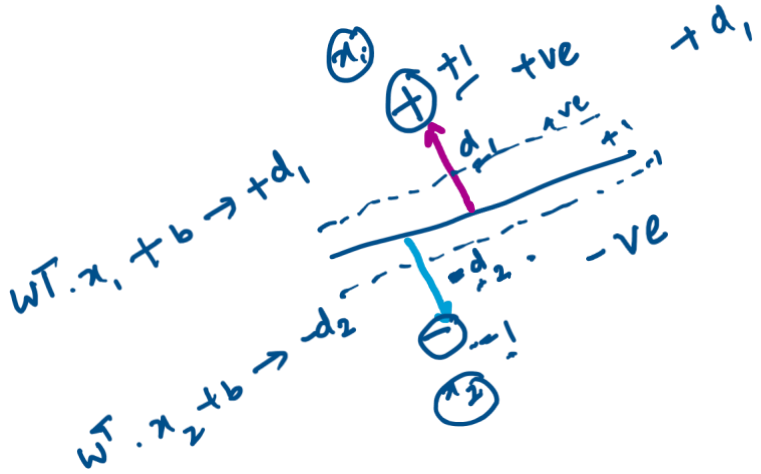
# Hard margin classifier




loss - '0'

$y_i^+ \rightarrow +1$

$y_i^- \rightarrow -1$

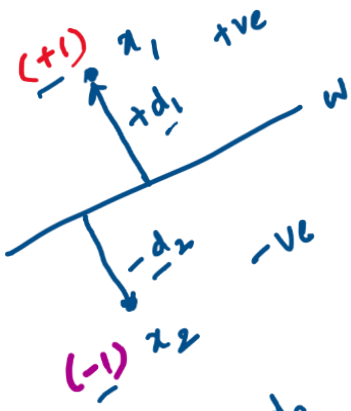



$y_i(w^T.x+b)$

$\pi \rightarrow \boxed{y_i(w^T.x+b)} > 0$ - correctly classified point

$\pi^+ \quad y_i(w^T.x+b) \geq 1$ ✓

$\pi^- \quad y_i(w^T.x+b) \geq 1$




$y_i(w^T.x+b) > 0 \quad w^T.x_1+b = +d_1 \quad w^T.x_2+b = -d_2$

$\pi^+ \quad \boxed{y_i(w^T.x+b) \geq 1}$

$\pi^- \quad \boxed{y_i(w^T.x+b) \leq -1}$

1 is the margin

**Loss** 'margin d' $\frac{2}{\|w\|}$ s.t

$$\underset{\vec{w}}{\arg\max} \frac{2}{\|w\|} \quad \text{s.t} \quad y_i(w^T.x + b) \geq 1$$

$$\underset{(\frac{1}{x})}{\min} \|w\|$$

**Goal** all points should be perfectly classified.

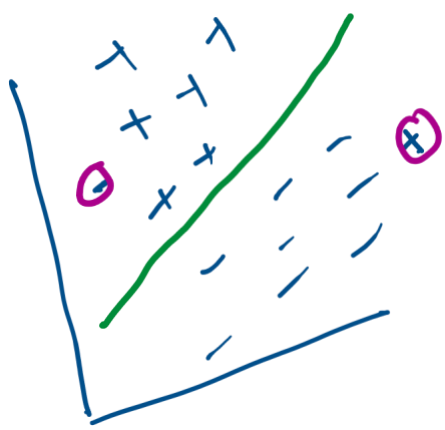
error - zero ✓

## Hard margin classifier

**loss**

$$\underset{\vec{w}}{\arg\max} \frac{2}{\|w\|} \quad \text{s.t} \quad y_i(w^T.x + b) \geq 1 \checkmark$$

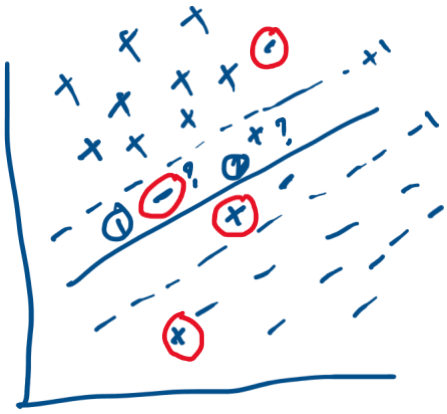
'zero'



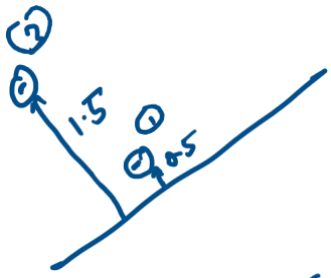
→ Hard margin classifier

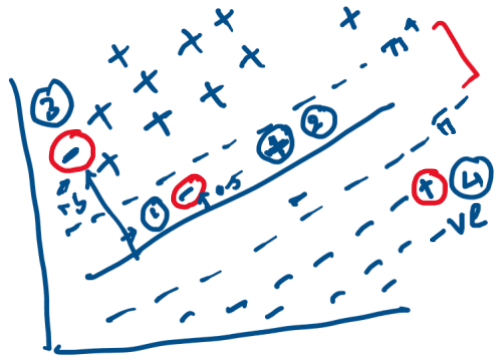
Hard margin classifier doesn't work if there is no linear separable classifier

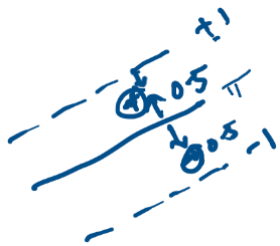
# Soft margin classifier



## loss




$$y_i(w^T \cdot x + b) = 0.5$$

$$= 1 - 0.5 = 0.5$$

$$-1 - (-0.5) = (-0.5)(-1) = 0.5$$

error → 'δ'

a +ve $x_1$ at 0.5 distance from $\pi$ and $\pi^+$

$\xi_1$ error → 0.5 ✓

a −ve $x_2$ at 0.5 distance from $\pi$ and $\pi^-$

$\xi_2$ error → 0.5

$\min \|w\|$ , $\quad \xi \; \delta \rightarrow$ <u>minimum</u>

$$y_i(w^T.x + b) \geq 1 - \xi_i \quad \sim 0.5$$

$\delta_i = 0$ for all correctly classified points

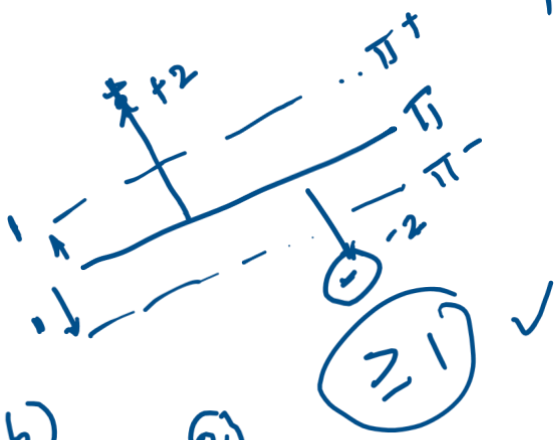
$\delta_i > 0$ for misclassified points

$$\underline{\text{Loss}} \; \underset{\vec{w}}{\text{argmin}} \; \frac{\|w\|}{2} + \frac{C}{N} \sum_{i=1}^{N} \xi_i \quad \text{S.T} \quad (w^T.x + b)\, y_i \geq 1 - \xi_i$$

$\boxed{\text{C - hypermater.}}$

→ <u>Loss</u> '<u>Hinge loss</u>' ① $y_i(w^T.x + b) \geq 1$ all correct predictions.

$\boxed{\text{Loss} = 0}$




$y_i = +1$

$$y_i \left( \frac{w^T.x + b}{+2} \right) \sim ②$$

$+1$ , $+2$ , $-2$

$-1$ $(-2)$

No penality

→ for correctly classified points no penality ✓

**Case-2**

Correct Classified but in the gutter

penality loss ~ $0 < y \cdot (w^T x + b) < 1$ $\boxed{\sim 0.5}$

③ **Case-3**

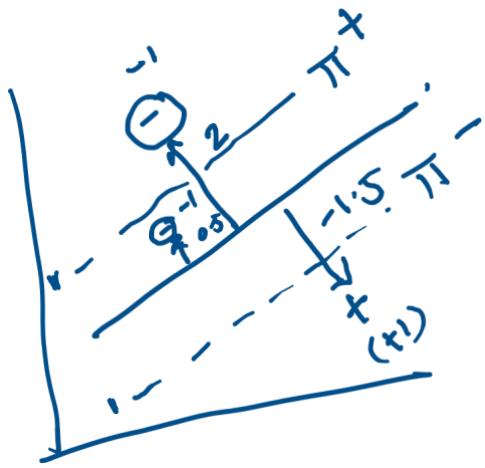
Wrong classification

$$\boxed{y \cdot (w^T \cdot x + b) \leq 0}$$

$-1 \cdot (2) \sim -2$

$\sim -0.5$

**Case-3** strong penality




$(-1.5)(+1) = -1.5$

**Case-1** ✓

Correctly classified

No penality

**Case-2** ✓

Correct classif

gutter

slight penality

**Case-3** ✓

wrongly clas

high penality

$$\mathcal{L} = \max\left(0,\ 1 - y_i(w^T.x + b)\right)$$

| | $y$ | $w^T.x+b$ | $y(w^T.x+b)$ | Loss |
|---|---|---|---|---|
| ① | $+1$ | $2$ | $2$ | $0$ ✓ |
| ② | $+1$ | $0.5$ | $0.5$ | $0.5$ |
| ③ | $+1$ | $-1$ | $-1$ | $2$ |
| | $-1$ | $0.5$ | $-0.5$ | $+1.5$ ✓ |

$$\mathcal{L} \rightarrow \text{Max}\left(0,\ 1 - y_i(w^T.x + b)\right)$$

$$(0,\ 1-2) \rightarrow \max(0, -1) \rightarrow$$

② $\max(0,\ 1 - 0.5)$ $\max(0, 0.5) \rightarrow 0.5$

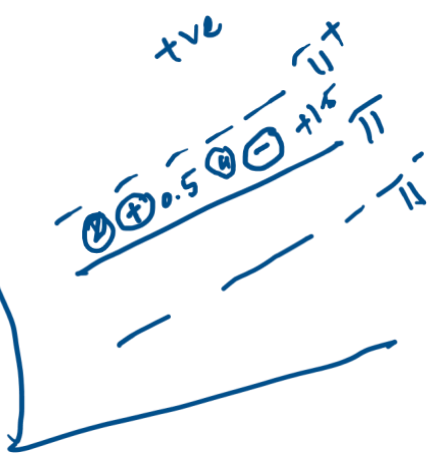
③ $\max(0, 1-(-1))$ $\max(0, 2)$ $- 2$

④ $\max(0,\ 1-(-0.5))$ $\max(0, 1.5) = 1.5$

Hinge loss

$$\text{Min} \rightarrow \frac{\|w\|}{2} + \boxed{C \cdot \frac{1}{N} \sum_{i=1}^{N} \delta_i}$$

$\underset{w^T}{\text{argmin}}$



C - control hyper parameter

loss fun + $\lambda$ $\|w\|$

Small value - we are okay more errors

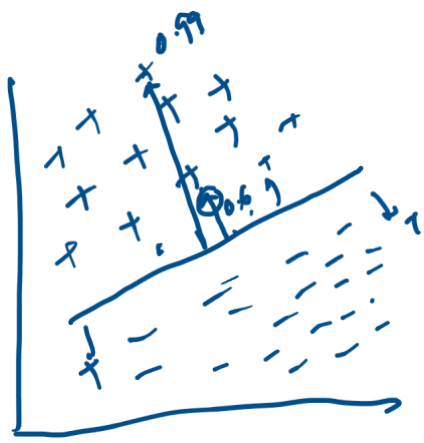
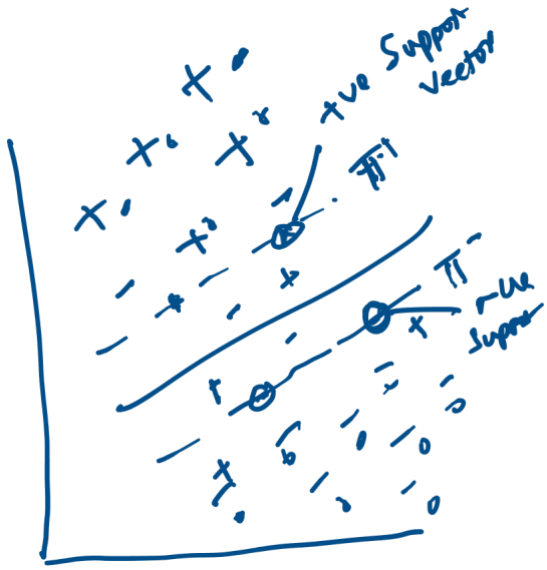
large value of C - we should have very less errors

Loss $\Rightarrow$ $\underset{w,b}{\arg\min} \dfrac{\|w\|}{2} + C \cdot \dfrac{1}{N} \sum_{i=1}^{N} \xi_i$ $\rightarrow$ Hinge loss.

$\xi_i$ $\max(0, 1 - y_i(w^T \cdot x + b))$

$$-\sum_{i=1}^{N} y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y})$$

$$\sum_{i=1}^{N} \max(0, 1 - z_i)$$

$z_i = y_i(w^T x + w_0)$

$y_i \begin{cases} -1 \\ +1 \end{cases}$ ?